

Nanyang Technological University
AY2024-25, Semester 1
MH2100 (Calculus III)
Midterm Exam 2

Date: October 24, 2024

Time allowed: 90 minutes

Full Name _____

Matriculation Number _____

Tutorial Group _____

INSTRUCTIONS TO CANDIDATES

- (1) **Do not turn over the exam paper until instructed to do so.**
- (2) This midterm exam consists of **only three** problems, each consisting of two parts.
- (3) Candidates are required to attempt **all three** problems.
- (4) The point allocation for each problem is shown at the top right-hand corner of its page.
- (5) Candidates may write anywhere on the blank pages reserved for showing work.
- (6) A **single** double-sided A4-sized help-sheet with only handwritten notes is permitted.

Problem 1	
Problem 2	
Problem 3	
Total	

Problem 1**Point allocation:** 20 points

- (a) Define a function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ by

$$f(x, y, z) \stackrel{\text{df}}{=} y^2 - 10z.$$

Let S denote the sphere with center $(0, 0, 0)$ and radius 6, i.e.,

$$S = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 36\}.$$

[10 points] Use the Method of Lagrange Multipliers to find the absolute extrema of $f|_S$.

Remark: $f|_S$ denotes the restriction of f to S .

- (b) Let D denote the closed disk in $\mathbb{R}^2$ with center $(0, 0)$ and radius 1, i.e.,

$$D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}.$$

Define a function $f : D \rightarrow \mathbb{R}$ by

$$f(x, y) \stackrel{\text{df}}{=} 2x^2 + y^2 - y + 3.$$

- (i) **[4 points]** Find all the critical points of f , as well as the values of f at these points.

Remark: Any critical point of f must belong to the interior of D .

- (ii) **[4 points]** Using the standard parametrization $\mathbf{r}$ of the unit circle, given by

$$\forall t \in [0, 2\pi] : \quad \mathbf{r}(t) = (\cos(t), \sin(t)),$$

find the absolute extrema of f on the boundary of D (which is the unit circle).

- (iii) **[2 points]** Use your work for (b)(i) and (b)(ii) to identify the absolute extrema of f .

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Problem 2**Point allocation:** 15 points

- (a) [5 points] Evaluate $\int_0^1 \left[\int_{\sqrt{y}}^1 \frac{1}{\sqrt{x^3+1}} dx \right] dy$.

Remark: First sketch the region of integration.

- (b) Let E denote the region in the first octant of $\mathbb{R}^3$ (i.e., $\mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0}$) that is

- bounded above by the sphere given by the equation $x^2 + y^2 + z^2 = 1$;
- bounded below by the closed cone given by the equation $z = \sqrt{\frac{x^2 + y^2}{3}}$.

- (i) [3 points] Describe E in spherical coordinates, using inequalities for ρ , θ , and ϕ .

- (ii) [7 points] Evaluate $\iiint_E 3z \, dV(x, y, z)$ using your work for (b)(i).

Caution: Work that does not use spherical coordinates **will not receive credit**.

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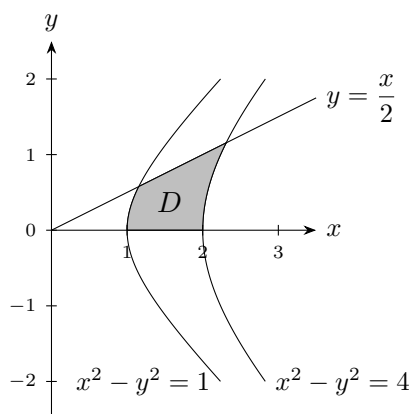
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Problem 3**Point allocation:** 15 points

(a) Let D denote the region in the first quadrant of $\mathbb{R}^2$ bounded by the following four curves:

- The two hyperbolas given by the equations $x^2 - y^2 = 1$ and $x^2 - y^2 = 4$.
- The two lines given by the equations $y = 0$ and $y = \frac{x}{2}$.



[7 points] Evaluate $\iint_D \frac{y}{x} \, dA(x, y)$ using a change of variables.

(b) (i) Define a function $f : \mathbb{R}^3 \setminus \{(0, 0, 0)\} \rightarrow \mathbb{R}$ by

$$f(x, y, z) \stackrel{\text{df}}{=} \frac{1}{x^2 + y^2 + z^2}.$$

Let C denote the curve in $\mathbb{R}^3$ having the parametrization $\mathbf{r} : [0, 2\pi] \rightarrow \mathbb{R}^3$ defined by

$$\mathbf{r}(t) \stackrel{\text{df}}{=} (\cos(t), \sin(t), t).$$

[4 points] Evaluate $\int_C f \, ds$.

(ii) Define a vector field $\mathbf{F}$ on $\mathbb{R}^3$ by

$$\mathbf{F}(x, y, z) \stackrel{\text{df}}{=} (x, 3xy, -(x + z)).$$

Let C denote the directed line segment going from $(1, 4, 2)$ to $(0, 5, 1)$.

[4 points] Evaluate $\int_C \mathbf{F} \bullet d\mathbf{r}$.

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